GES-207-4 — Mathematical Methods in Physics

Lecture 06

Hilbert Spaces: Completeness, L^2, and Infinite Orthonormal Bases

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Dr. José Paulo Marchezi — INPE

Learning goals

This lecture closes the first module by extending inner products, norms, orthogonality, projection, and orthonormal bases to infinite-dimensional spaces. By the end of these notes, you should be able to:

1. distinguish convergence from the Cauchy property;
2. explain completeness and define a Hilbert space;
3. explain why $C([a,b])$ with the L^2 norm is not complete;
4. define $L^2([a,b])$ and equality almost everywhere;
5. work with the sequence space ℓ^2;
6. interpret infinite orthonormal expansions through partial sums;
7. derive Bessel's inequality and Parseval's identity;
8. distinguish L^2 convergence from pointwise convergence;
9. connect Hilbert-space geometry with Fourier series.

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1 The central question

In a finite-dimensional space, an expansion in a basis is a finite sum. In an infinite-dimensional space, it becomes a sequence of partial sums. Before writing

$$f = \sum_{k=1}^{\infty} c_k u_k,$$

we must answer three separate questions:

1. In which norm should the partial sums converge?
2. Do the coefficients produce a Cauchy sequence of partial sums?
3. If they do, does the limit still belong to the original space?

The third question leads to completeness. Completeness is what prevents a well-behaved sequence from converging toward an object missing from the space.

Guiding chain of ideas

partial sums $\Longrightarrow$ Cauchy sequences $\Longrightarrow$ completeness
$\Longrightarrow$ infinite expansions whose limits remain in the space

2 From finite to infinite dimensions

Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be an orthonormal basis of $\mathbb{R}^n$ or $\mathbb{C}^n$. Every vector satisfies

$$\mathbf{v} = \sum_{k=1}^{n} \langle \mathbf{e}_k, \mathbf{v} \rangle \, \mathbf{e}_k$$

and

$$\|\mathbf{v}\|^2 = \sum_{k=1}^{n} | \langle \mathbf{e}_k, \mathbf{v} \rangle |^2.$$

The first formula decomposes a vector into components; the second is the Pythagorean theorem.

In infinite dimensions, the analogous expression is a series,

$$f = \sum_{k=1}^{\infty} c_k u_k.$$

The symbol on the right means the limit of the partial sums

$$S_N = \sum_{k=1}^{N} c_k u_k.$$

We must specify both whether S_N converges and whether its limit belongs to the space under consideration.

3 Norm convergence and Cauchy sequences

Convergence in norm

A sequence $\{v_n\}$ in a normed space V converges to $v \in V$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad \|v_n - v\| < \varepsilon.$$

We write $v_n \to v$.

For example, in $\mathbb{R}^2$,

$$v_n = \begin{pmatrix} 1 + 1/n \\ 2 - 1/n \end{pmatrix} \longrightarrow v = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

because

$$\|v_n - v\| = \sqrt{\frac{1}{n^2} + \frac{1}{n^2}} = \frac{\sqrt{2}}{n} \longrightarrow 0.$$

Cauchy sequence

A sequence $\{v_n\} \subset V$ is a **Cauchy sequence** if

$$\boxed{\forall \varepsilon > 0,\ \exists N \in \mathbb{N} : m, n \geq N \Longrightarrow \|v_m - v_n\| < \varepsilon.}$$

Convergence compares v_n with a known candidate v. The Cauchy condition compares late terms with one another and does not require the limit to be known in advance.

Every convergent sequence is Cauchy. Indeed, if $v_n \to v$, then

$$\begin{aligned} \|v_m - v_n\| &= \|(v_m - v) + (v - v_n)\| \\ &\leq \|v_m - v\| + \|v_n - v\|. \end{aligned}$$

For sufficiently large m, n, each term on the right is smaller than $\varepsilon/2$, so $\|v_m - v_n\| < \varepsilon$.

The converse is not always true. Whether every Cauchy sequence converges depends on the space.

4 Completeness and Hilbert spaces

Complete space

A normed space V is **complete** if every Cauchy sequence in V converges in the norm of V to an element that also belongs to V.

Hilbert space

An inner-product space $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ is a **Hilbert space** if it is complete in the induced norm

$$\|f\| = \sqrt{\langle f, f \rangle}.$$

Every finite-dimensional normed space over $\mathbb{R}$ or $\mathbb{C}$ is complete. Completeness becomes a genuinely new issue in infinite dimensions.

The familiar inclusion $\mathbb{Q} \subset \mathbb{R}$ provides an analogy. A sequence of rational numbers can be Cauchy and converge to $\sqrt{2}$, which is missing from $\mathbb{Q}$. The real numbers fill that gap. In a related sense,

$$C([a, b]) \longrightarrow L^2([a, b])$$

is a completion when $C([a, b])$ is equipped with the L^2 norm.

5 A noncomplete function space

On $C([0,1])$, define

$$\langle f, g\rangle = \int_0^1 \overline{f(x)} g(x)\, dx, \qquad \|f\|_2 = \left(\int_0^1 |f(x)|^2\, dx\right)^{1/2}.$$

This is a valid inner product and norm on continuous functions, but the space is not complete in this norm.

Example 6.1: continuous functions with a discontinuous L^2 limit

For $n \geq 2$, define

$$f_n(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{2} - \frac{1}{n}, \\ n\left(x - \frac{1}{2} + \frac{1}{n}\right), & \frac{1}{2} - \frac{1}{n} < x < \frac{1}{2}, \\ 1, & \frac{1}{2} \leq x \leq 1. \end{cases}$$

Every f_n is continuous. The natural limit is the step function

$$H(x) = \begin{cases} 0, & x < 1/2, \\ 1, & x \geq 1/2, \end{cases}$$

which is not continuous at $x = 1/2$.
The functions differ only on the transition interval. Setting $y = x - \frac{1}{2} + \frac{1}{n}$,

$$\begin{aligned} \|f_n - H\|_2^2 &= \int_{\frac{1}{2}-\frac{1}{n}}^{\frac{1}{2}} n^2 \left(x - \frac{1}{2} + \frac{1}{n}\right)^2 dx \\ &= n^2 \int_0^{1/n} y^2 dy = \frac{1}{3n}. \end{aligned}$$

Therefore,

$$\|f_n - H\|_2 = \frac{1}{\sqrt{3n}} \longrightarrow 0.$$

The sequence is Cauchy in the L^2 norm, but its limit is not in $C([0,1])$. Hence $C([0,1])$ is not complete in this norm.

The failure lies in the space, not in the sequence: the space of continuous functions is too small to contain a perfectly meaningful L^2 limit.

6 The space L^2 and equality almost everywhere

The space $L^2([a,b])$

$L^2([a,b])$ is the space of equivalence classes of measurable functions $f : [a,b] \to \mathbb{K}$, with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, such that

$$\boxed{\int_a^b |f(x)|^2 dx < \infty.}$$

Its inner product and norm are

$$\boxed{\langle f, g\rangle = \int_a^b \overline{f(x)} g(x) dx}, \qquad \boxed{\|f\|_2 = \left(\int_a^b |f(x)|^2 dx\right)^{1/2}}.$$

The fundamental result is that $L^2([a,b])$ is complete; therefore, it is a Hilbert space. Its full proof requires tools from measure theory. For our purposes, the conclusion is essential: every L^2-Cauchy sequence has an L^2 limit.

Why do we use equivalence classes? Let $f(x) = 0$ everywhere and let g equal 1 at $x = 1/2$ and 0 elsewhere. Then

$$\|f - g\|_2^2 = \int_0^1 |f(x) - g(x)|^2 dx = 0.$$

A norm cannot assign zero distance to distinct vectors. We therefore identify functions that differ only on a set of measure zero.

Equality almost everywhere

Functions f and g are equal **almost everywhere**, abbreviated a.e., if $f(x) = g(x)$ except possibly on a set of measure zero. In L^2, functions equal almost everywhere represent the same element.

Example 6.2: a discontinuous function in L^2

For the step function H above,

$$\int_0^1 |H(x)|^2 dx = \int_{1/2}^1 1\, dx = \frac{1}{2}.$$

Thus $H \in L^2([0,1])$. Continuity is not required.

Example 6.3: an integrable singularity

Let $f(x) = x^{-1/4}$ on $(0,1]$. Although $f(x) \to \infty$ as $x \to 0^+$,

$$\int_0^1 |f(x)|^2 dx = \int_0^1 x^{-1/2} dx = 2.$$

Therefore, $x^{-1/4} \in L^2((0,1))$.

Example 6.4: a function outside L^2

For $g(x) = x^{-1/2}$,

$$\int_0^1 |g(x)|^2 dx = \int_0^1 \frac{dx}{x} = \infty.$$

Hence $x^{-1/2} \notin L^2((0,1))$.

7 The sequence space ℓ^2

The Hilbert space ℓ^2

$$\ell^2 = \left\{ a = (a_1, a_2, \ldots) : \sum_{k=1}^{\infty} |a_k|^2 < \infty \right\}.$$

The inner product and norm are

$$\langle a, b \rangle = \sum_{k=1}^{\infty} \overline{a_k} b_k, \qquad \|a\|_{\ell^2} = \left(\sum_{k=1}^{\infty} |a_k|^2 \right)^{1/2}.$$

The space ℓ^2 is complete.

For the canonical vectors $e_1 = (1, 0, 0, \ldots)$, $e_2 = (0, 1, 0, \ldots)$, and so on,

$$\langle e_j, e_k \rangle = \delta_{jk}.$$

Every $a = (a_1, a_2, \ldots) \in \ell^2$ satisfies

$$a = \sum_{k=1}^{\infty} a_k e_k, \qquad \|a\|_{\ell^2}^2 = \sum_{k=1}^{\infty} |a_k|^2.$$

This is the model for orthonormal expansions in L^2: the coefficients of a function form a square-summable sequence.

8 Infinite orthonormal systems and bases

Orthonormal system

A sequence $\{u_k\}_{k=1}^{\infty}$ is orthonormal if

$$\boxed{\langle u_j, u_k \rangle = \delta_{jk}.}$$

Thus $\|u_k\| = 1$ and $u_j \perp u_k$ for $j \neq k$.

Orthonormality alone does not guarantee that all vectors can be represented. The system must also be complete.

Orthonormal basis of a Hilbert space

An orthonormal system $\{u_k\}$ is an **orthonormal basis**, or a complete orthonormal system, if every $f \in \mathcal{H}$ satisfies

$$\boxed{f = \sum_{k=1}^{\infty} \langle u_k, f \rangle \, u_k} \qquad \text{with convergence in norm.}$$

Equivalently,

$$\overline{\operatorname{span}\{u_1, u_2, \ldots\}} = \mathcal{H},$$

or the only vector orthogonal to every u_k is the zero vector.

The coefficients retain the finite-dimensional projection formula:

$$\boxed{c_k = \langle u_k, f \rangle .}$$

The equality $f = \sum_k c_k u_k$ means precisely that

$$\|f - S_N\| \longrightarrow 0, \qquad S_N = \sum_{k=1}^{N} c_k u_k.$$

9 Finite projections, Bessel, and Parseval

Let

$$S_N = \sum_{k=1}^{N} \langle u_k, f \rangle \, u_k, \qquad V_N = \operatorname{span}\{u_1, \ldots, u_N\}.$$

Then S_N is the orthogonal projection of f onto V_N. If $r_N = f - S_N$, then, for $j \leq N$,

$$\begin{aligned} \langle u_j, r_N \rangle &= \langle u_j, f \rangle - \sum_{k=1}^{N} \langle u_k, f \rangle \, \langle u_j, u_k \rangle \\ &= 0. \end{aligned}$$

Thus $r_N \perp V_N$. The Pythagorean theorem gives

$$\boxed{\|f\|^2 = \sum_{k=1}^{N} |\langle u_k, f \rangle|^2 + \|f - S_N\|^2 .} \tag{1}$$

Since the last term is nonnegative, taking $N \to \infty$ gives Bessel's inequality.

Bessel's inequality

For any orthonormal system and every $f \in \mathcal{H}$,

$$\boxed{\sum_{k=1}^{\infty} |\langle u_k, f \rangle|^2 \leq \|f\|^2 .}$$

This does not require the orthonormal system to be complete.

Bessel's inequality shows that the coefficient sequence belongs to ℓ^2. If the system is complete, then $S_N \to f$, so the error term in (1) tends to zero.

Parseval's identity

If $\{u_k\}$ is an orthonormal basis of $\mathcal{H}$, then

$$\boxed{f = \sum_{k=1}^{\infty} \langle u_k, f \rangle \, u_k} \quad \text{in norm,}$$

and

$$\boxed{\|f\|^2 = \sum_{k=1}^{\infty} |\langle u_k, f \rangle|^2.}$$

Parseval is the Pythagorean theorem in infinitely many dimensions. The geometry is unchanged; what is new is the need to control convergence.

10 Complex exponentials and Fourier coefficients

On $[0,1]$, define

$$u_k(x) = e^{2\pi i k x}, \qquad k \in \mathbb{Z}.$$

Their norms are

$$\|u_k\|_2^2 = \int_0^1 e^{-2\pi i k x} e^{2\pi i k x} dx = 1.$$

For $j \neq k$,

$$\begin{aligned}\langle u_j, u_k \rangle &= \int_0^1 e^{2\pi i (k-j)x} dx \\ &= \left[\frac{e^{2\pi i (k-j)x}}{2\pi i (k-j)} \right]_0^1 = 0,\end{aligned}$$

because $e^{2\pi i (k-j)} = 1$. Hence

$$\langle u_j, u_k \rangle = \delta_{jk}.$$

In fact, $\{e^{2\pi i k x}\}_{k \in \mathbb{Z}}$ is a complete orthonormal system in $L^2([0,1])$. Therefore,

$$f(x) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k x} \quad \text{in } L^2,$$

where

$$\boxed{c_k = \int_0^1 e^{-2\pi i k x} f(x) dx.}$$

Parseval becomes

$$\boxed{\int_0^1 |f(x)|^2 dx = \sum_{k \in \mathbb{Z}} |c_k|^2.}$$

It says that the total squared norm is preserved when we pass from the function $f(x)$ to its sequence of Fourier coefficients.

Example 6.5: Fourier coefficients of $f(x) = x$

For $k = 0$,

$$c_0 = \int_0^1 x \, dx = \frac{1}{2}.$$

For $k \neq 0$, integration by parts gives

$$\begin{aligned}c_k &= \int_0^1 x e^{-2\pi i k x} dx \\ &= \left[\frac{x e^{-2\pi i k x}}{-2\pi i k} \right]_0^1 - \int_0^1 \frac{e^{-2\pi i k x}}{-2\pi i k} dx \\ &= -\frac{1}{2\pi i k} = \frac{i}{2\pi k}.\end{aligned}$$

Thus, in $L^2([0,1])$,

$$x = \frac{1}{2} + \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{i}{2\pi k} e^{2\pi i k x}.$$

Parseval checks the result:

$$\int_0^1 x^2 dx = \frac{1}{4} + 2\sum_{k=1}^{\infty} \frac{1}{4\pi^2 k^2}$$
$$= \frac{1}{4} + \frac{1}{2\pi^2}\frac{\pi^2}{6} = \frac{1}{3}.$$

11 L^2 convergence is not pointwise convergence

For partial sums S_N, convergence in L^2 means

$$\boxed{\|S_N - f\|_2 = \left(\int |S_N(x) - f(x)|^2 dx\right)^{1/2} \longrightarrow 0.}$$

This does not automatically imply $S_N(x) \to f(x)$ at every point.

Pointwise convergence

f_n converges pointwise to f if, for every fixed x,

$$f_n(x) \longrightarrow f(x).$$

Pointwise convergence examines each point separately; L^2 convergence examines the integrated squared error. Questions of pointwise convergence of Fourier series require additional hypotheses and additional theorems.

For example, let $f_n(x) = x^n$ on $[0, 1]$. Pointwise, $f_n(x) \to 0$ for $x < 1$, while $f_n(1) = 1$. Nevertheless,

$$\|f_n\|_2^2 = \int_0^1 x^{2n} dx = \frac{1}{2n+1} \longrightarrow 0.$$

Thus $f_n \to 0$ in L^2. The exceptional value at one point is invisible in L^2 because a single point has measure zero.

12 Best approximation and the role of completeness

The projection

$$S_N = \sum_{k=1}^{N} \langle u_k, f\rangle\, u_k$$

is the best approximation to f from $V_N = \text{span}\{u_1, \ldots, u_N\}$. For every $g \in V_N$,

$$f - g = (f - S_N) + (S_N - g),$$

and the two terms are orthogonal. Therefore,

$$\|f - g\|^2 = \|f - S_N\|^2 + \|S_N - g\|^2 \geq \|f - S_N\|^2 .$$

Hence

$$\boxed{\|f - S_N\| \leq \|f - g\| .}$$

Now consider a formal orthonormal series

$$\sum_{k=1}^{\infty} c_k u_k.$$

For $m > n$,

$$\|S_m - S_n\|^2 = \sum_{k=n+1}^{m} |c_k|^2.$$

If $\sum_k |c_k|^2 < \infty$, the tails tend to zero and $\{S_N\}$ is Cauchy. Completeness then guarantees a limit $f \in \mathcal{H}$.

Why completeness is essential

$$\boxed{\begin{array}{c}\text{square-summable coefficients} \Longrightarrow \text{Cauchy partial sums} \\ \Longrightarrow \text{completeness} \Longrightarrow \text{a limit inside } \mathcal{H}\end{array}}$$

Without completeness, the partial sums could be Cauchy while pointing toward an object outside the space.

An orthonormal Hilbert basis should not be confused with a Hamel basis. A Hamel basis represents each vector by a finite linear combination. A Hilbert basis permits norm-convergent infinite expansions.

13 Conceptual summary

Concept	Meaning
Cauchy sequence	Late terms become arbitrarily close to one another.
Completeness	Every Cauchy sequence has a limit inside the space.
Hilbert space	A complete inner-product space.
$L^2([a,b])$	Equivalence classes of square-integrable functions.
Equality a.e.	Changes on measure-zero sets do not change the L^2 element.
Orthonormal basis	A complete orthonormal system with norm-convergent expansions.
Bessel	An orthonormal system captures no more than the total squared norm.
Parseval	A complete system captures exactly the total squared norm.
L^2 convergence	Integrated squared error tends to zero.

Common pitfalls

- A Cauchy sequence need not converge unless the space is complete.
- L^2 elements are equivalence classes, not distinguished by values on measure-zero sets.
- Orthonormality does not imply completeness of the system.
- Bessel's inequality holds for any orthonormal system; Parseval's identity requires an

orthonormal basis.

- Convergence in L^2 does not automatically imply pointwise or uniform convergence.
- An infinite orthonormal expansion means convergence of partial sums in norm, not an algebraic sum with infinitely many terms.

Core formulas

$$\langle f, g\rangle_{L^2} = \int_a^b \overline{f(x)} g(x) dx, \qquad \|f\|_2^2 = \int_a^b |f(x)|^2 dx.$$

$$c_k = \langle u_k, f\rangle, \qquad S_N = \sum_{k=1}^{N} \langle u_k, f\rangle u_k.$$

$$\sum_{k=1}^{\infty} |\langle u_k, f\rangle|^2 \le \|f\|^2 \qquad \text{(Bessel)}.$$

$$f = \sum_{k=1}^{\infty} \langle u_k, f\rangle u_k, \qquad \|f\|^2 = \sum_{k=1}^{\infty} |\langle u_k, f\rangle|^2 \qquad \text{(Parseval)}.$$

14 Take-home assignment: five problems

Special Thursday assignment — due Thursday, October 15, 2026

Complete all five problems and submit one organized solution set at the beginning of class on **Thursday, October 15, 2026**. Show intermediate steps, state the results you use, and explain your reasoning in complete sentences when requested. Answers without supporting work are incomplete.
No solutions are included in these notes.

Problem 1 — Cauchy sequences and completeness in ℓ^2

For each $n \in \mathbb{N}$, define

$$a^{(n)} = \left(1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, 0, 0, \ldots\right).$$

(a) For $m > n$, write $a^{(m)} - a^{(n)}$ explicitly.

(b) Show that

$$\left\|a^{(m)} - a^{(n)}\right\|_{\ell^2}^2 = \sum_{k=n+1}^{m} \frac{1}{k^2}.$$

(c) Use convergence of $\sum_{k=1}^{\infty} k^{-2}$ to prove that $\{a^{(n)}\}$ is Cauchy.

(d) Identify its natural limit a and prove that $a \in \ell^2$.

(e) Prove directly that $a^{(n)} \to a$ in ℓ^2 and explain where completeness enters conceptually.

Problem 2 — Membership in L^2

For real α, define $f_\alpha(x) = x^{-\alpha}$ on $(0, 1)$.

(a) Determine all α for which $f_\alpha \in L^1((0,1))$.

(b) Determine all α for which $f_\alpha \in L^2((0,1))$.

(c) Give an explicit value of α for which $f_\alpha \in L^1$ but $f_\alpha \notin L^2$.

(d) For every admissible α, compute $\|f_\alpha\|_2$.

(e) Explain why the value assigned to $f_\alpha(0)$ does not affect its element of L^2.

Problem 3 — Projection, Bessel, and best approximation

In $L^2([-1, 1])$, use the orthonormal functions

$$u_0(x) = \frac{1}{\sqrt{2}}, \qquad u_1(x) = \sqrt{\frac{3}{2}}\, x.$$

Let $f(x) = x^2 + x$.

(a) Verify that u_0 and u_1 are orthonormal.

(b) Compute $c_0 = \langle u_0, f\rangle$ and $c_1 = \langle u_1, f\rangle$.

(c) Construct the projection $S_1 = c_0 u_0 + c_1 u_1$.

(d) Compute $\|f\|^2$, $|c_0|^2 + |c_1|^2$, and $\|f - S_1\|^2$, and verify the finite Pythagorean identity.

(e) Explain why S_1 is the best approximation to f among all linear functions $a + bx$.

Problem 4 — Fourier coefficients and Parseval

Let $f(x) = e^{2\pi i x} + 2e^{-4\pi i x}$ on $[0, 1]$.

(a) Compute every Fourier coefficient

$$c_k = \int_0^1 e^{-2\pi i k x} f(x) dx.$$

(b) Write the Fourier expansion of f and identify all nonzero modes.

(c) Compute $\int_0^1 |f(x)|^2 dx$ directly.

(d) Compute $\sum_{k \in \mathbb{Z}} |c_k|^2$ and verify Parseval's identity.

(e) Interpret the result as preservation of squared norm between the physical-space and coefficient-space representations.

Problem 5 — Comparing modes of convergence

On $[0, 1]$, define

$$g_n(x) = \sqrt{n}\, \mathbf{1}_{[0,1/n]}(x),$$

where $\mathbf{1}_E$ is the indicator function of the set E.

(a) Determine the pointwise limit of $g_n(x)$ for every $x \in [0, 1]$.

(b) Compute $\|g_n\|_2$.

(c) Decide whether $g_n \to 0$ in $L^2([0, 1])$ and justify the answer.

(d) Compute $\|g_n - g_m\|_2^2$ for $m > n$ and determine whether $\{g_n\}$ is a Cauchy sequence in L^2.

(e) Explain clearly what this example shows about the difference between pointwise convergence and L^2 convergence.